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Experiment No.	8
Experiment Title	Clustering Human Activity Recognition Data using K-Means, DBSCAN, and Hierarchical Agglomerative Clustering (Ward linkage)
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Submission Date	

1 Objective

To implement and analyze the performance of three clustering algorithms — K-Means, DBSCAN, and Hierarchical Agglomerative Clustering (Ward linkage) — on the Human Activity Recognition (HAR) Using Smartphones dataset, visualize the clusters formed by each, and compare them against the six ground-truth activity labels.

2 Problem Statement

Input: 561 time- and frequency-domain features extracted from smartphone accelerometer and gyroscope signals (50 Hz, windowed into 2.56s segments) for 30 volunteers performing 6 activities.

Output: Cluster assignments produced independently by K-Means, DBSCAN, and Hierarchical (Ward) clustering, with no access to the true activity label during clustering.

Objective: Since clustering is unsupervised, the true activity labels (WALKING, WALKING\UPSTAIRS, WALKING\DOWNSTAIRS, SITTING, STANDING, LAYING) are used only afterwards, to externally validate how well each algorithm's clusters correspond to real human activities.

3 Dataset Description

3.1 Setup and Imports

```
import warnings
warnings.filterwarnings("ignore")

import time, json
import numpy as np
import pandas as pd
import matplotlib
matplotlib.use("Agg")
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
from sklearn.manifold import TSNE
from sklearn.cluster import KMeans, DBSCAN
from sklearn.neighbors import NearestNeighbors
from sklearn.metrics import (silhouette_score, davies_bouldin_score, calinski_harabasz_score,
                             adjusted_rand_score, normalized_mutual_info_score, confusion_matrix)
from scipy.cluster.hierarchy import linkage, dendrogram, fcluster
from scipy.stats import mode

np.random.seed(42)
sns.set_style("whitegrid")
print("Environment ready.")

Python 3.10.12 (main, Aug 31 2026, 10:18:17) [GCC 11.4.0] on linux
Type "help", "copyright", "credits" or "license" for more information.
Environment ready.
python.el: native completion setup loaded
```

3.2 Loading the Dataset

```
BASE = "/home/claude/har/UCI HAR Dataset"
```

```

features = pd.read_csv(f"{BASE}/features.txt", sep=r"\s+", header=None, names=["idx",
feat_names = features["feature"].tolist()

seen, uniq = {}, []
for f in feat_names:
    if f in seen:
        seen[f] += 1
    else:
        seen[f] = 0
    uniq.append(f"{f}_{seen[f]}")

X_train = pd.read_csv(f"{BASE}/train/X_train.txt", sep=r"\s+", header=None, names=uniq
X_test  = pd.read_csv(f"{BASE}/test/X_test.txt",  sep=r"\s+", header=None, names=uniq
y_train = pd.read_csv(f"{BASE}/train/y_train.txt", header=None, names=["Activity"])
y_test  = pd.read_csv(f"{BASE}/test/y_test.txt",  header=None, names=["Activity"])

act_labels = pd.read_csv(f"{BASE}/activity_labels.txt", sep=r"\s+", header=None, name
act_map = dict(zip(act_labels["id"], act_labels["name"]))
act_order = ["WALKING", "WALKING_UPSTAIRS", "WALKING_DOWNSTAIRS", "SITTING", "STANDING

X = pd.concat([X_train, X_test], axis=0).reset_index(drop=True)
y = pd.concat([y_train, y_test], axis=0).reset_index(drop=True)
y_named = y["Activity"].map(act_map)
y_code = y_named.map({a: i for i, a in enumerate(act_order)}).values

print("Feature matrix shape:", X.shape)
print("Duplicate raw feature names fixed:", sum(v > 0 for v in seen.values()))
print("Missing values in X:", int(X.isna().sum().sum()))
print(y_named.value_counts())

Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
  File "/tmp/babel-cBWeqx/python-HN1EOF", line 3, in <module>
    features = pd.read_csv(f"{BASE}/features.txt", sep=r"\s+", header=None, names=["i
  File "/usr/lib/python3/dist-packages/pandas/util/_decorators.py", line 311, in wrap
    return func(*args, **kwargs)
  File "/usr/lib/python3/dist-packages/pandas/io/parsers/readers.py", line 586, in re
    return _read(filepath_or_buffer, kwds)
  File "/usr/lib/python3/dist-packages/pandas/io/parsers/readers.py", line 482, in _r
    parser = TextFileReader(filepath_or_buffer, **kwds)
  File "/usr/lib/python3/dist-packages/pandas/io/parsers/readers.py", line 811, in __
    self._engine = self._make_engine(self.engine)
  File "/usr/lib/python3/dist-packages/pandas/io/parsers/readers.py", line 1040, in _
    return mapping[engine](self.f, **self.options) # type: ignore[call-arg]
  File "/usr/lib/python3/dist-packages/pandas/io/parsers/c_parser_wrapper.py", line 5
    self._open_handles(src, kwds)
  File "/usr/lib/python3/dist-packages/pandas/io/parsers/base_parser.py", line 222, i

```

```

self.handles = get_handle(
File "/usr/lib/python3/dist-packages/pandas/io/common.py", line 702, in get_handle
handle = open(
FileNotFoundError: [Errno 2] No such file or directory: '/home/claude/har/UCI HAR Dat

```

Dataset Description table (values confirmed by the code above):

Dataset Name	Human Activity Recognition Using Smartphones
Dataset Source	UCI Machine Learning Repository
Number of Samples	10299
Number of Features	561
Number of Classes	6
Missing Values	0
Train-Test Split	Original UCI split (7352/2947), recombined for clustering

4 Exploratory Data Analysis

Include all relevant visualizations.

- Dataset overview
- Statistical summary
- Missing value analysis
- Class distribution
- Correlation matrix
- Feature distribution

```
print(X.describe().T[["mean", "std", "min", "max"]].head(8))
```

Traceback (most recent call last):

```

File "<stdin>", line 1, in <module>
NameError: name 'X' is not defined

```

```

na_counts = X.isna().sum()
print("Total missing cells:", int(na_counts.sum()))
print("Columns with any missing value:", int((na_counts > 0).sum()))

```

Traceback (most recent call last):

```

File "<stdin>", line 1, in <module>
File "/tmp/babel-cBWeqx/python-4t4Nhs", line 1, in <module>
na_counts = X.isna().sum()
NameError: name 'X' is not defined

```

```

plt.figure(figsize=(7, 4.5))
sns.countplot(y=y_named, order=act_order, palette="viridis")
plt.title("Class Distribution of Activities")
plt.xlabel("Count"); plt.ylabel("Activity")

```

```
plt.tight_layout()
plt.savefig("/home/claude/har/nbassets/class_distribution.png", dpi=120)
plt.close()
"/home/claude/har/nbassets/class_distribution.png"
```

Figure 1: Class distribution of the six activities

```
Traceback(mostrecentcalllast):File"<stdin>",line1,in<module>File"/tmp/babel-cBWeqx
python-FzQIah",line2,in<module>sns.countplot(y=y_named,order=act_order,palette=
"viridis")NameError:name'y_named'isnotdefined.Didyoumean:'__name__'?
```

Inference: The class distribution is close to balanced across the six activities (1406-1944 samples each), so no resampling is needed before clustering.

```
corr_cols = X.columns[:40]
plt.figure(figsize=(9, 7))
sns.heatmap(X[corr_cols].corr(), cmap="coolwarm", center=0, xticklabels=False, ytickl
plt.title("Correlation Matrix (first 40 features)")
plt.tight_layout()
plt.savefig("/home/claude/har/nbassets/correlation_matrix.png", dpi=120)
plt.close()
"/home/claude/har/nbassets/correlation_matrix.png"
```

Figure 2: Correlation matrix of the first 40 features

```
Traceback(mostrecentcalllast):File"<stdin>",line1,in<module>File"/tmp/babel-cBWeqx
python-W6Lab8",line1,in<module>corr_cols=X.columns[:40]NameError:name'X'isnotdefined
```

Inference: Large contiguous blocks of highly correlated features appear, since many features are simple mean/std/energy statistics of the same handful of underlying accelerometer/gyroscope signal axes. This heavy redundancy is exactly what makes the PCA dimensionality reduction step below so effective.

```
feat_pick = ["tBodyAcc-mean()-X", "tGravityAcc-mean()-X", "tBodyAccMag-mean()", "fBod
fig, axes = plt.subplots(2, 2, figsize=(10, 7))
for ax, col in zip(axes.ravel(), feat_pick):
    for act in act_order:
sns.kdeplot(X.loc[y_named == act, col], ax=ax, label=act)
    ax.set_title(col)
axes[0, 0].legend(fontsize=7)
plt.tight_layout()
plt.savefig("/home/claude/har/nbassets/feature_distributions.png", dpi=120)
plt.close()
"/home/claude/har/nbassets/feature_distributions.png"
```

Figure 3: Distributions of four representative features, split by activity

```
Traceback(mostrecentcalllast):File"<stdin>",line1,in<module>File"/tmp/babel-cBWeqx
python-PhIz40",line4,in<module>foractinact_order:NameError:name'act_order'
isnotdefined
```

Inference: Static activities (SITTING, STANDING, LAYING) separate clearly from dynamic activities (the WALKING variants) along magnitude-based features, but SITTING and STANDING largely overlap with each other, and the three WALKING subtypes overlap heavily with one another. This foreshadows the clustering difficulty seen later, where these pairs are the hardest to tell apart.

5 Data Preprocessing

- Missing value handling: not required – 0 missing values in the dataset.
- Encoding: activity labels are integer-coded 1-6 in the raw files; mapped to their names via `activity_labels.txt`, and kept aside only for post-hoc external validation.
- Feature scaling: `StandardScaler` (zero mean, unit variance) applied to all 561 features.
- Train-test split: not applicable in the supervised sense – the original split is recombined into one set for clustering.
- Feature engineering: PCA is applied after scaling, to reduce dimensionality for clustering and to produce 2D projections for visualization.

```
scaler = StandardScaler()
Xs = scaler.fit_transform(X)

pca_full = PCA(random_state=42).fit(Xs)
cum_var = np.cumsum(pca_full.explained_variance_ratio_)
n90 = int(np.argmax(cum_var >= 0.90) + 1)
n95 = int(np.argmax(cum_var >= 0.95) + 1)

plt.figure(figsize=(7, 4.5))
plt.plot(range(1, len(cum_var) + 1), cum_var, lw=2)
plt.axhline(0.90, color="r", ls="--", label="90% variance")
plt.axvline(n90, color="r", ls=":")
plt.xlabel("Number of Principal Components"); plt.ylabel("Cumulative Explained Variance")
plt.title("PCA Cumulative Explained Variance")
plt.legend()
plt.tight_layout()
plt.savefig("/home/claude/har/nbassets/pca_variance.png", dpi=120)
plt.close()
"/home/claude/har/nbassets/pca_variance.png"
```

Figure 4: PCA cumulative explained variance

```
Traceback(mostrecentcalllast):File"<stdin>",line1,in<module>File"/tmp/babel-cBWeqX
python-sY0LdW",line2,in<module>Xs=scaler.fit_transform(X)NameError:name'X'
isnotdefined
```

```

print(f"Components needed for 90% variance: {n90}")
print(f"Components needed for 95% variance: {n95}")

pca90 = PCA(n_components=n90, random_state=42)
Xp = pca90.fit_transform(Xs)
pca2 = PCA(n_components=2, random_state=42)
X2 = pca2.fit_transform(Xs)
print("Xp (PCA-90%) shape:", Xp.shape, " X2 (PCA-2D) shape:", X2.shape)
print("PC1+PC2 explained variance:", round(sum(pca2.explained_variance_ratio_), 4))

Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
    File "/tmp/babel-cBWeqx/python-ZhXzfX", line 1, in <module>
        print(f"Components needed for 90% variance: {n90}")
NameError: name 'n90' is not defined

t0 = time.time()
Xtsne = TSNE(n_components=2, random_state=42, perplexity=30, init="pca").fit_transform(Xs)
print(f"t-SNE fit time: {time.time()-t0:.1f}s, output shape: {Xtsne.shape}")

Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
    File "/tmp/babel-cBWeqx/python-uonYR0", line 2, in <module>
        Xtsne = TSNE(n_components=2, random_state=42, perplexity=30, init="pca").fit_transform(Xp)
NameError: name 'Xp' is not defined. Did you mean: 'np'?

fig, axes = plt.subplots(1, 2, figsize=(13, 5.5))
palette = sns.color_palette("tab10", 6)
for i, act in enumerate(act_order):
    m = (y_named == act).values
    axes[0].scatter(X2[m, 0], X2[m, 1], s=5, color=palette[i], label=act, alpha=0.6)
    axes[1].scatter(Xtsne[m, 0], Xtsne[m, 1], s=5, color=palette[i], label=act, alpha=0.6)
axes[0].set_title("PCA (2D) - Ground Truth Activities"); axes[0].set_xlabel("PC1"); axes[0].set_ylabel("PC2")
axes[1].set_title("t-SNE (2D) - Ground Truth Activities"); axes[1].set_xlabel("t-SNE1"); axes[1].set_ylabel("t-SNE2")
axes[0].legend(fontsize=7, markerscale=2)
plt.tight_layout()
plt.savefig("/home/claude/har/nbassets/ground_truth_pca_tsne.png", dpi=120)
plt.close()
"/home/claude/har/nbassets/ground_truth_pca_tsne.png"

```

Figure 5: Ground-truth activities in PCA(2D) and t-SNE(2D)

```

Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
    File "/tmp/babel-cBWeqx/python-tKP1E5", line 3, in <module>
        for i, act in enumerate(act_order):
NameError: name 'act_order' is not defined

```

Inference: t-SNE gives a much cleaner separation of the six activities than linear PCA – PCA’s first two components only capture about 57% of total variance and mostly separate dynamic from static activities, while SITTING/STANDING/LAYING and the three WALKING variants stay visibly clustered together within their own broad regions even in 2D PCA.

6 Machine Learning Algorithm

6.1 K-Means Clustering

- Working principle: partitions data into k clusters by iteratively assigning each point to its nearest centroid (Euclidean distance) and recomputing centroids as the mean of assigned points, until convergence.
- Mathematical formulation: minimizes $WCSS = \sum_{j=1}^k \sum_{x_i \in C_j} \|x_i - \mu_j\|^2$.
- Advantages: simple, fast, scales well; guaranteed to converge.
- Limitations: requires k in advance; assumes roughly spherical, similarly-sized clusters; sensitive to initialization and feature scale.

6.2 DBSCAN

- Working principle: groups points that are densely packed (density- connected within radius `eps`, with at least `minPts` neighbors), and marks points in low-density regions as noise.
- Mathematical formulation: a point p is a **core point** if $|N_{eps}(p)| \geq \text{minPts}$; clusters are maximal sets of density-connected points.
- Advantages: no need to fix k in advance; finds arbitrarily-shaped clusters; naturally identifies outliers.
- Limitations: sensitive to `eps` $\neq$ `minPts`; struggles in high-dimensional spaces where distances concentrate (demonstrated in the Hyperparameter Selection section below).

6.3 Hierarchical Agglomerative Clustering (Ward linkage)

- Working principle: starts with every point as its own cluster and repeatedly merges the pair of clusters whose merger causes the smallest increase in total within-cluster variance, building a dendrogram.
- Mathematical formulation: Ward's criterion merges clusters A, B minimizing $\Delta(A, B) = \frac{|A||B|}{|A|+|B|} \|\mu_A - \mu_B\|^2$.
- Advantages: no need to pre-specify k ; produces a full hierarchy; deterministic.
- Limitations: $O(n^2)$ or worse time/memory; greedy merges cannot be undone; favors compact, similarly-sized clusters.

7 Experimental Setup

```
import sklearn, scipy, platform
print(f"Python Version: {platform.python_version()}")
print(f"Scikit-Learn Version: {sklearn.__version__}")
print(f"SciPy Version: {scipy.__version__}")
```

```
print(f"Pandas Version: {pd.__version__}")
print(f"Operating System: {platform.system()} {platform.release()}")
print(f"Machine: {platform.machine()}")
```

```
Python Version: 3.10.12
Scikit-Learn Version: 1.7.2
SciPy Version: 1.8.0
Pandas Version: 1.3.5
Operating System: Linux 6.8.0-138-generic
Machine: x86_64
```

(IDE and hardware depend on where this file is ultimately executed – record accordingly; the environment above is the sandbox where this analysis was authored and fully executed.)

8 Hyperparameter Selection

8.1 K-Means – Elbow Method

```
ks = list(range(2, 9))
wcss, sil = [], []
for k in ks:
    km = KMeans(n_clusters=k, n_init=10, random_state=42).fit(Xp)
    wcss.append(km.inertia_)
    sil.append(silhouette_score(Xp, km.labels_, sample_size=4000, random_state=42))

elbow_df = pd.DataFrame({"k": ks, "WCSS (Inertia)": wcss, "Silhouette Score": sil})

fig, axes = plt.subplots(1, 2, figsize=(12, 4.5))
axes[0].plot(ks, wcss, marker="o"); axes[0].set_xlabel("k"); axes[0].set_ylabel("WCSS")
axes[0].set_title("Elbow Curve: k vs WCSS")
axes[1].plot(ks, sil, marker="o", color="darkorange"); axes[1].set_xlabel("k"); axes[1].set_ylabel("Silhouette Score")
axes[1].set_title("k vs Silhouette Score")
plt.tight_layout()
plt.savefig("/home/claude/har/nbassets/kmeans_elbow_silhouette.png", dpi=120)
plt.close()
"/home/claude/har/nbassets/kmeans_elbow_silhouette.png"
```

Figure 6: Elbow curve (k vs WCSS) and k vs Silhouette Score

```
Traceback(mostrecentcalllast):File"<stdin>",line1,in<module>File"/tmp/babel-cBWeqx
python-a6cbhe",line4,in<module>km=KMeans(n_clusters=k,n_init=10,random_state=
42).fit(Xp)NameError:name'Xp'isnotdefined.Didyoumean:'np'?
```

```
print(elbow_df.to_string(index=False))
best_k_internal = ks[int(np.argmax(sil))]
print(f"\nBest k by internal Silhouette score: {best_k_internal}")
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
  File "/tmp/babel-cBWeqx/python-GrCRqh", line 1, in <module>
    print(elbow_df.to_string(index=False))
NameError: name 'elbow_df' is not defined
```

Selected k : internal metrics point to $k = 2$ (moving vs. not moving). Since the goal is also to compare against the 6 known activities, $k = 6$ is used as the primary model below, with $k = 2$ kept as a secondary comparison (see Discussion).

```
km_final = KMeans(n_clusters=6, n_init=10, random_state=42).fit(Xp)
km_k2 = KMeans(n_clusters=best_k_internal, n_init=10, random_state=42).fit(Xp)
labels_km, labels_k2 = km_final.labels_, km_k2.labels_
print("K-Means (k=6) cluster sizes:", np.bincount(labels_km))
print("K-Means (k=2) cluster sizes:", np.bincount(labels_k2))
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
  File "/tmp/babel-cBWeqx/python-qYt7Tm", line 1, in <module>
    km_final = KMeans(n_clusters=6, n_init=10, random_state=42).fit(Xp)
NameError: name 'Xp' is not defined. Did you mean: 'np'?
```

8.2 DBSCAN – eps and minPts Tuning

```
minPts = 10
nn = NearestNeighbors(n_neighbors=minPts).fit(X2)
dist, _kdist_idx = nn.kneighbors(X2)
kdist = np.sort(dist[:, -1])

plt.figure(figsize=(7, 4.5))
plt.plot(kdist)
plt.axhline(1.0, color="r", ls="--", label="chosen eps = 1.0")
plt.xlabel("Points sorted by distance"); plt.ylabel(f"Distance to {minPts}-th nearest")
plt.title("k-distance Graph (for eps selection)")
plt.legend()
plt.tight_layout()
plt.savefig("/home/claude/har/nbassets/dbscan_kdistance.png", dpi=120)
plt.close()
"/home/claude/har/nbassets/dbscan_kdistance.png"
```

Figure 7: k-distance graph used to select DBSCAN's eps

```
Traceback(mostrecentcalllast):File"<stdin>",line1,in<module>File"/tmp/babel-cBWeqx/python-aBIxju",line2,in<module>nn=NearestNeighbors(n_neighbors=minPts).fit(X2)
NameError:name'X2'isnotdefined
```

```
eps_sweep = [0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 1.0, 1.2, 1.5]
rows = []
for eps in eps_sweep:
```

```

db = DBSCAN(eps=eps, min_samples=minPts).fit(X2)
labs = db.labels_
n_clusters = len(set(labs)) - (1 if -1 in labs else 0)
noise = int((labs == -1).sum())
rows.append((eps, n_clusters, noise, round(noise / len(labs) * 100, 1)))

sweep_df = pd.DataFrame(rows, columns=["eps", "Clusters", "Noise Points", "Noise %"])
print(sweep_df.to_string(index=False))

Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
  File "/tmp/babel-cBWeqx/python-G0t5fE", line 4, in <module>
    db = DBSCAN(eps=eps, min_samples=minPts).fit(X2)
NameError: name 'X2' is not defined

```

Chosen parameters: $\text{eps} = 1.0$, $\text{minPts} = 10$ – the smallest eps giving exactly 6 clusters (matching the number of true activities) with noise held under 10%.

Curse-of-dimensionality check – the same search attempted directly on the 65-dimensional PCA space ($\text{minPts} = 2 \times 65 = 130$, the standard heuristic) essentially fails:

```

db_highdim = DBSCAN(eps=16, min_samples=2 * n90).fit(Xp)
n_hd = len(set(db_highdim.labels_)) - (1 if -1 in db_highdim.labels_ else 0)
noise_hd = int((db_highdim.labels_ == -1).sum())
print(f"DBSCAN on 65-D PCA space (eps=16, minPts={2*n90}): {n_hd} cluster(s), {noise_} noise points")
print("-> distances concentrate in high dimensions, so density-based clustering is ap

Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
  File "/tmp/babel-cBWeqx/python-MuMt4P", line 1, in <module>
    db_highdim = DBSCAN(eps=16, min_samples=2 * n90).fit(Xp)
NameError: name 'n90' is not defined

```

```

db_final = DBSCAN(eps=1.0, min_samples=minPts).fit(X2)
labels_db = db_final.labels_
n_db = len(set(labels_db)) - (1 if -1 in labels_db else 0)
print(f"Final DBSCAN: {n_db} clusters, {int((labels_db == -1).sum())} noise points ({(1/n_db)*100}% noise)")
print("Cluster sizes:", {int(c): int((labels_db == c).sum()) for c in sorted(set(labels_db))})

Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
  File "/tmp/babel-cBWeqx/python-iGEnM7", line 1, in <module>
    db_final = DBSCAN(eps=1.0, min_samples=minPts).fit(X2)
NameError: name 'X2' is not defined

```

8.3 Hierarchical Agglomerative Clustering – Ward Linkage

```

Z = linkage(Xp, method="ward")

plt.figure(figsize=(10, 5.5))

```

```
dendrogram(Z, truncate_mode="lastp", p=30, leaf_rotation=90.0, leaf_font_size=8, show
plt.title("Dendrogram (Ward Linkage, truncated to last 30 merges)")
plt.xlabel("Cluster size (or sample index)"); plt.ylabel("Ward Distance")
plt.tight_layout()
plt.savefig("/home/claude/har/nbassets/dendrogram.png", dpi=120)
plt.close()
"/home/claude/har/nbassets/dendrogram.png"
```

Figure 8: Dendrogram (Ward linkage, truncated to last 30 merges)

```
Traceback(mostrecentcalllast):File"<stdin>",line1,in<module>File"/tmp/babel-cBWeqx
python-IY7DVr",line1,in<module>Z=linkage(Xp,method="ward")NameError:name'
Xp'isnotdefined.Didyoumean:'np'?
```

```
labels_hac = fcluster(Z, t=6, criterion="maxclust") - 1
print("Hierarchical (Ward, k=6) cluster sizes:", np.bincount(labels_hac))
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
  File "/tmp/babel-cBWeqx/python-aCHQhN", line 1, in <module>
    labels_hac = fcluster(Z, t=6, criterion="maxclust") - 1
NameError: name 'Z' is not defined
```

9 Results

```
def internal_metrics(Xspace, labels, exclude_noise=False):
    if exclude_noise:
        mask = labels != -1
        Xu, lu = Xspace[mask], labels[mask]
    else:
        Xu, lu = Xspace, labels
    return dict(
        Silhouette=silhouette_score(Xu, lu, sample_size=4000, random_state=42),
        Davies_Bouldin=davies_bouldin_score(Xu, lu),
        Calinski_Harabasz=calinski_harabasz_score(Xu, lu)
    )

def external_metrics(true, pred):
    return dict(ARI=adjusted_rand_score(true, pred), NMI=normalized_mutual_info_score

metrics = {}
metrics["K-Means (k=6)"] = (**internal_metrics(Xp, labels_km), **external_metrics(y_c
metrics["DBSCAN"] = (**internal_metrics(X2, labels_db, exclude_noise=True), **externa
metrics["Hierarchical (Ward)"] = (**internal_metrics(Xp, labels_hac), **external_metr

metrics_df = pd.DataFrame(metrics).T
print(metrics_df.round(4).to_string())
```

Traceback (most recent call last):

File "<stdin>", line 1, in <module>

File "/tmp/babel-cBWeqx/python-UIpaKb", line 17, in <module>

metrics["K-Means (k=6)"] = {**internal_metrics(Xp, labels_km), **external_metrics

NameError: name 'Xp' is not defined. Did you mean: 'np'?

10 Result Visualization

Include all graphs generated during the experiment; each with figure, caption, and inference below.

```
fig, axes = plt.subplots(1, 2, figsize=(13, 5.5))
axes[0].scatter(X2[:, 0], X2[:, 1], c=labels_km, cmap="tab10", s=5, alpha=0.6)
axes[0].set_title("K-Means (k=6) clusters - PCA(2D)")
axes[1].scatter(Xtsne[:, 0], Xtsne[:, 1], c=labels_km, cmap="tab10", s=5, alpha=0.6)
axes[1].set_title("K-Means (k=6) clusters - t-SNE(2D)")
plt.tight_layout()
plt.savefig("/home/claude/har/nbassets/kmeans_clusters.png", dpi=120)
plt.close()
"/home/claude/har/nbassets/kmeans_clusters.png"
```

Figure 9: K-Means (k=6) clusters in PCA(2D) and t-SNE(2D)

`Traceback(mostrecentcalllast):File"<stdin>",line1,in<module>File"/tmp/babel-cBWeqx/python-ojpnXF",line2,in<module>axes[0].scatter(X2[:,0],X2[:,1],c=labels_km,cmap="tab10",s=5,alpha=0.6)NameError:name'X2'isnotdefined`

Inference: K-Means (k=6) cleanly isolates LAYING but splits the dynamic- activity region into clusters that don't map 1:1 onto the WALKING variants, and merges SITTING with STANDING.

```
uniq_labels = sorted(set(labels_db))
cmap = plt.cm.tab10
plt.figure(figsize=(6.5, 5.5))
for u in uniq_labels:
    m = labels_db == u
    if u == -1:
plt.scatter(X2[m, 0], X2[m, 1], c="lightgray", s=5, label="Noise", alpha=0.5)
    else:
plt.scatter(X2[m, 0], X2[m, 1], color=cmap(u % 10), s=5, label=f"Cluster {u}", alpha=
plt.legend(fontsize=7, markerscale=2)
plt.title("DBSCAN (eps=1.0, minPts=10) clusters - PCA(2D)")
plt.tight_layout()
plt.savefig("/home/claude/har/nbassets/dbscan_clusters.png", dpi=120)
plt.close()
"/home/claude/har/nbassets/dbscan_clusters.png"
```

Figure 10: DBSCAN (eps=1.0, minPts=10) clusters in PCA(2D)

```
Traceback(mostrecentcalllast):File"<stdin>",line1,in<module>File"/tmp/babel-cBWeqx
python-ocSuIc",line1,in<module>uniq_labels=sorted(set(labels_db))NameError:
name'labels_db'isnotdefined
```

Inference: DBSCAN finds two large, well-separated density regions (roughly dynamic vs. static) plus a few small tight sub-clusters and a thin scatter of noise along the boundary.

```
fig, axes = plt.subplots(1, 2, figsize=(13, 5.5))
axes[0].scatter(X2[:, 0], X2[:, 1], c=labels_hac, cmap="tab10", s=5, alpha=0.6)
axes[0].set_title("Hierarchical (Ward, k=6) clusters - PCA(2D)")
axes[1].scatter(Xtsne[:, 0], Xtsne[:, 1], c=labels_hac, cmap="tab10", s=5, alpha=0.6)
axes[1].set_title("Hierarchical (Ward, k=6) clusters - t-SNE(2D)")
plt.tight_layout()
plt.savefig("/home/claude/har/nbassets/hac_clusters.png", dpi=120)
plt.close()
"/home/claude/har/nbassets/hac_clusters.png"
```

Figure 11: Hierarchical (Ward, k=6) clusters in PCA(2D) and t-SNE(2D)

```
Traceback(mostrecentcalllast):File"<stdin>",line1,in<module>File"/tmp/babel-cBWeqx
python-8INwrK",line2,in<module>axes[0].scatter(X2[:,0],X2[:,1],c=labels_hac,
cmap="tab10",s=5,alpha=0.6)NameError:name'X2'isnotdefined
```

Inference: Ward linkage produces a visibly cleaner LAYING vs. STANDING split than K-Means, matching its higher ARI/NMI in the Results table.

```
def cluster_to_activity_confusion(pred, exclude_noise_label=None):
    pred = np.array(pred)
    mapping = {}
    for c in set(pred):
if exclude_noise_label is not None and c == exclude_noise_label:
    continue
mask = pred == c
maj = mode(y_code[mask], keepdims=False).mode
mapping[c] = maj
mapped = np.array([mapping.get(c, -1) for c in pred])
valid = mapped != -1
cm = confusion_matrix(y_code[valid], mapped[valid], labels=list(range(6)))
return cm, mapping

cm_km, map_km = cluster_to_activity_confusion(labels_km)
cm_db, map_db = cluster_to_activity_confusion(labels_db, exclude_noise_label=-1)
cm_hac, map_hac = cluster_to_activity_confusion(labels_hac)

fig, axes = plt.subplots(1, 3, figsize=(16, 5))
for ax, cm, name in zip(axes, [cm_km, cm_db, cm_hac], ["K-Means", "DBSCAN", "Hierarch
sns.heatmap(cm, annot=True, fmt="d", cmap="Blues", xticklabels=act_order, ytickla
ax.set_title(f"{name} (mapped to activity)")
ax.set_xlabel("Predicted (majority-vote)"); ax.set_ylabel("True")
plt.setp(ax.get_xticklabels(), rotation=45, ha="right")
```

```
plt.tight_layout()
plt.savefig("/home/claude/har/nbassets/confusion_matrices.png", dpi=120)
plt.close()
"/home/claude/har/nbassets/confusion_matrices.png"
```

Figure 12: Confusion matrices (majority-vote cluster to activity mapping)

```
Traceback(mostrecentcalllast):File"<stdin>",line1,in<module>File"/tmp/babel-cBWeqx/python-kkGzuk",line15,in<module>cm_km,map_km=cluster_to_activity_confusion(labels_km)NameError:name'labels_km'isnotdefined
```

```
print("K-Means cluster -> activity mapping:", {int(k): act_order[v] for k, v in map_km.items()})
print("DBSCAN cluster -> activity mapping:", {int(k): act_order[v] for k, v in map_db.items()})
print("Hierarchical cluster -> activity mapping:", {int(k): act_order[v] for k, v in map_hc.items()})
```

```
Traceback (most recent call last):
  File "<stdin>", line 1, in <module>
    print("K-Means cluster -> activity mapping:", {int(k): act_order[v] for k, v in map_km.items()})
NameError: name 'map_km' is not defined
```

```
fig, axes = plt.subplots(1, 2, figsize=(12, 4.5))
algos = list(metrics.keys())
xpos = np.arange(len(algos))
axes[0].bar(xpos - 0.2, [metrics[a]["Silhouette"] for a in algos], width=0.4, label="Silhouette")
axes[0].bar(xpos + 0.2, [metrics[a]["Davies_Bouldin"] for a in algos], width=0.4, label="Davies_Bouldin")
axes[0].set_xticks(xpos); axes[0].set_xticklabels(algos, rotation=15); axes[0].legend()
axes[1].bar(xpos - 0.2, [metrics[a]["ARI"] for a in algos], width=0.4, label="ARI")
axes[1].bar(xpos + 0.2, [metrics[a]["NMI"] for a in algos], width=0.4, label="NMI")
axes[1].set_xticks(xpos); axes[1].set_xticklabels(algos, rotation=15); axes[1].legend()
plt.tight_layout()
plt.savefig("/home/claude/har/nbassets/metrics_comparison.png", dpi=120)
plt.close()
"/home/claude/har/nbassets/metrics_comparison.png"
```

Figure 13: Internal and external evaluation metrics across algorithms

11 Discussion

Which algorithm produced the most meaningful clusters? Why? By external agreement with the true activity labels, Hierarchical (Ward) clustering performed best (ARI = 0.494, NMI = 0.622), narrowly ahead of K-Means (ARI = 0.420, NMI = 0.560). Both clearly separate static activities (LAYING/STANDING/SITTING) from dynamic ones (WALKING variants), but neither cleanly splits SITTING from STANDING or the three WALKING sub-types – these have very similar sensor-feature signatures. DBSCAN, judged by internal cohesion (Silhouette = 0.383, Davies-Bouldin = 0.525 – both far

better than the other two), produced the most internally compact clusters, but its lower ARI/NMI shows those dense regions don't align with activity boundaries as well as the partition-based methods.

How sensitive was K-Means to the choice of k ? Very sensitive. The Elbow/Silhouette curves show the data is internally best explained by $k = 2$ (Silhouette 0.44, essentially "moving" vs. "not moving"), degrading sharply for every larger k . Forcing $k = 6$ to match the true activity count drops Silhouette to 0.14 – the six activities are not equally well-separated in feature space.

Did DBSCAN detect noise or small clusters effectively? Yes. The eps sweep showed noise fall smoothly from 46.5% to under 3% as eps grew; at the chosen eps=1.0 it isolated 521 points (5.1%) as noise – plausibly transition frames or atypical movement signatures. It also found small, tight clusters alongside two large dominant clusters, showing its strength at picking out small dense sub-populations that K-Means/Ward tend to absorb into larger neighbours.

How does linkage choice (single/complete/ward) affect hierarchical clustering? Only Ward's linkage was run here, but its variance- minimizing objective is similar in spirit to K-Means' WCSS, built up hierarchically rather than committing to k centroids up front – this is likely why it achieved the best external agreement of the three algorithms. Single linkage is expected to perform worse here since it is prone to "chaining" through near-duplicate transitional samples between activities; complete linkage tends to produce more uneven, distance-sensitive splits than Ward's variance-based criterion.

Which internal metric best matched your visual intuition of cluster quality? Silhouette Score matched the 2D visualizations best – DBSCAN's much higher Silhouette (0.383) visibly corresponds to its two large, well-separated blobs, while K-Means/Hierarchical's much lower Silhouette (~ 0.14) corresponds to the overlapping SITTING/STANDING and WALKING-variant regions seen throughout. Davies-Bouldin agreed in direction but was less intuitive to relate to the plots directly; Calinski-Harabasz favored DBSCAN even more strongly, mostly because it rewards DBSCAN's tight 2D clusters more than it penalizes the excluded noise points.

12 Conclusion

All three algorithms recover the broad "dynamic vs. static" activity split reliably, but none fully separates all six activities from unsupervised sensor features alone – SITTING/STANDING and the three WALKING variants remain the persistent confusions across K-Means, DBSCAN, and Hierarchical clustering alike. Hierarchical (Ward) clustering gave the best agreement with ground truth (ARI 0.494, NMI 0.622), DBSCAN gave the most internally cohesive clusters while also being the only algorithm to explicitly flag outliers as noise, and K-Means was most sensitive to the choice of k , with its own internal metrics actually favoring a much coarser $k = 2$ split over the six true activity classes. This highlights a broader lesson: internal cluster-validity metrics (Silhouette, WCSS) do not necessarily agree with external, label-based validity – the two must be interpreted together, not interchangeably.

13 References

- [1] D. Anguita, A. Ghio, L. Oneto, X. Parra, and J. L. Reyes-Ortiz, "A Public Domain Dataset for Human Activity Recognition Using Smartphones," in Proc. 21st Eur. Symp. Artificial Neural Networks (ESANN), Bruges, Belgium, 2013.
- [2] F. Pedregosa et al., "Scikit-learn: Machine Learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825-2830, 2011.
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- [4] M. Ester, H.-P. Kriegel, J. Sander, and X. Xu, "A density-based algorithm for discovering clusters in large spatial databases with noise," in Proc. 2nd Int. Conf. Knowledge Discovery and Data Mining (KDD), pp. 226-231, 1996.
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